Collision system and energy dependence of di-hadron correlations

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Main physics points

- Jet-like correlation
  - No collision system dependence
  - Reasonably well described by PYTHIA
  - Perhaps some indication of modification in central Au+Au
- Ridge
  - Increases with system size
  - Jet/ridge ratio independent of collision energy (for these energies)

v_2

- Consistent with Oana and Joern's analyses
  - Appendices D and E in the analysis note
- Consistent $<v_2>$ with Marco
  - Using Marco's parameterization of v2(pT), centrality
  - He uses $<v_2^{trig}\, v_2^{assoc}>$, we use $v_2(<p_T^{trig}>)\, v_2(<p_T^{assoc}>)$
  - He weighs 0-5%, 5-10%, and 10-20% bins by percentage of cross section to get v_2 in 0-12%, we use the number of particles in these bins
  - Marco's paper does not correct for acceptance in $\Delta\eta$
  - Systematic errors: Marco assumes v_2^{trig} and v_2^{assoc} are not correlated and we assume 100% correlation.
    - Our assumption is more conservative but leads to larger systematic errors
- Cu+Cu systematic errors reasonable?

ZYAM

- We recognize ZYAM's limitations and have tried to admit to them in the text of the paper
- These limitations do not prevent us from extracting physics from the data
- Note that the jet-like correlation is not dependent on ZYAM, so only the data on the ridge are sensitive to this
- Other methods also have (different) limitations

2D Fits

- Lanny requested 2D fits and with extensive help from Chanaka DeSilva and Anthony Timmins, we were able to do these fits
- The interpretation is not straightforward. We are still thinking about these fits.
- These use $\cos(\Delta\varphi)$ to describe the away-side
  - Clearly not valid in Mach cone region
    - Note Mach cone seen in some PHENIX data before background subtraction
  - Clearly not valid in Punch Through region
- Background is lower in fits
- v_2 is higher
- Jet-like yield is consistent

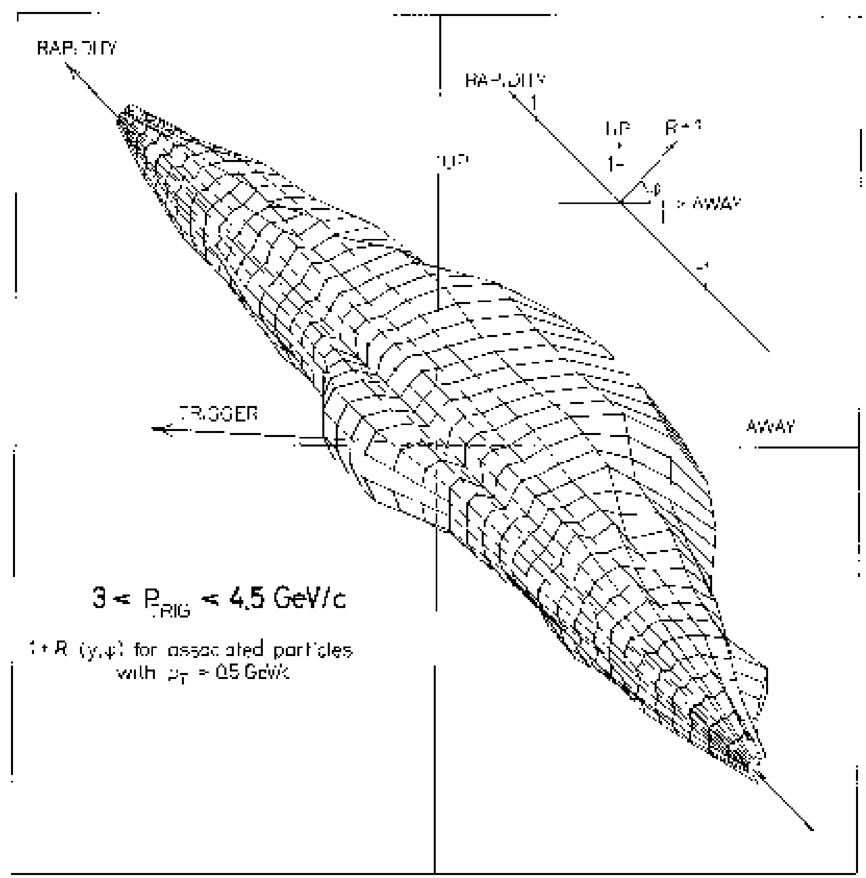
Areas open to improvement

- Should the more peripheral Au+Au 62 data be shown?
  - v_2 bins are wide, probably leading to overestimate of v_2
  - As a result, the error bar includes negative numbers. This is not physical.
- Extend discussion of limitations of ZYAM?
- Need N_{bin} for Au+Au@62 GeV
- Extend discussion of the models?
- Mention CMS, ISR p+p results in discussion?

Backup

ISR data

- Mike Tannenbaum has claimed ISR saw the ridge in p+p before CMS (http://arxiv.org/abs/1010.0964) and therefore before STAR
- This is the “evidence” (Nucl. Phys. B 145 (1978) 305)



ISR data

- This ISR figure is not background subtracted. In the absence of any correlations a cylindrical background would be observed.
- Fig. 6 shows the raw correlations relative to min. bias (no high p_T trigger) collisions. While there is a signal above 1 on the near-side, this is below the signal perpendicular to the trigger, showing a suppression – not an enhancement – of the yield on the near-side. This is more indicative of momentum conservation than a ridge, since a signal above 1 can be interpreted simply as more particles in an event with a high p_T particle.
- In the Au+Au 62 GeV data, the most central data have a statistical significance of 4σ with 800k central events. With 23k events – the ISR sample – the Au+Au 62 data would have a significance of 0.8σ. The ridge should be smaller in p+p and smaller at 200 GeV. STAR sees no evidence of a ridge in 200 GeV p+p with much higher statistics so it is not clear how ISR would see a ridge with 23k p+p events at 62 GeV.
- There is no discussion of a ridge in the ISR paper.
- It is Christine's strong opinion that the ISR data do not show a ridge. Whether and how to address this in the paper is an open issue.